Typing Rules and Pi Injectivity for Dependent Type Theory with $\mathsf{U} : \mathsf{U}$

1 Judgments

We work with three mutually defined judgment forms:

$\Gamma\ \mathsf{ctx}$	Γ is a well-formed context
$\Gamma \vdash M : A$	M has type A in context Γ
$\Gamma \vdash M = N : A$	M and N are definitionally equal at type A

Contexts are snoc-lists: $\Gamma ::= \cdot \mid \Gamma, x{:}A$.

2 Well-formed contexts

$$\frac{}{\cdot\ \mathsf{ctx}} \qquad\qquad \frac{\Gamma \vdash A : \mathsf{U}}{\Gamma, x{:}A\ \mathsf{ctx}}$$

3 Typing rules

$$\textsc{Var}\ \frac{\Gamma\ \mathsf{ctx} \qquad (x{:}A) \in \Gamma}{\Gamma \vdash x : A} \qquad \textsc{Conv}\ \frac{\Gamma \vdash M : A \qquad \Gamma \vdash A = B : \mathsf{U} \qquad \Gamma \vdash B : \mathsf{U}}{\Gamma \vdash M : B} \qquad \mathrm{U}\ \frac{\Gamma\ \mathsf{ctx}}{\Gamma \vdash \mathsf{U} : \mathsf{U}}$$

$$\textsc{Pi}\ \frac{\Gamma \vdash A : \mathsf{U} \qquad \Gamma, x{:}A \vdash B : \mathsf{U}}{\Gamma \vdash \Pi(x{:}A)B : \mathsf{U}} \qquad \textsc{Lam}\ \frac{\Gamma \vdash A : \mathsf{U} \qquad \Gamma, x{:}A \vdash B : \mathsf{U} \qquad \Gamma, x{:}A \vdash M : B}{\Gamma \vdash \lambda(x{:}A)M : \Pi(x{:}A)B}$$

$$\textsc{App}\ \frac{\Gamma \vdash A : \mathsf{U} \qquad \Gamma, x{:}A \vdash B : \mathsf{U} \qquad \Gamma \vdash f : \Pi(x{:}A)B \qquad \Gamma \vdash a : A}{\Gamma \vdash f\ a : B[a]}$$

4 Conversion rules

$$\textsc{Refl}\quad \frac{\Gamma \vdash M : A}{\Gamma \vdash M = M : A} \qquad \textsc{Sym}\quad \frac{\Gamma \vdash M = N : A}{\Gamma \vdash N = M : A} \qquad \textsc{Trans}\quad \frac{\Gamma \vdash M = N : A \qquad \Gamma \vdash N = P : A}{\Gamma \vdash M = P : A}$$

$$\textsc{Conv-Eq}\quad \frac{\Gamma \vdash M = N : A \qquad \Gamma \vdash A = B : \mathsf{U} \qquad \Gamma \vdash B : \mathsf{U}}{\Gamma \vdash M = N : B}$$

$$\textsc{Beta}\quad \frac{\Gamma \vdash A : \mathsf{U} \qquad \Gamma, x{:}A \vdash B : \mathsf{U} \qquad \Gamma, x{:}A \vdash M : B \qquad \Gamma \vdash a : A}{\Gamma \vdash (\lambda(x{:}A)M)\ a = M[a] : B[a]}$$

$$\textsc{Pi-Cong}\quad \frac{\Gamma \vdash A = A' : \mathsf{U} \qquad \Gamma, x{:}A \vdash B = B' : \mathsf{U}}{\Gamma \vdash \Pi(x{:}A)B = \Pi(x{:}A')B' : \mathsf{U}}$$

$$\textsc{Funext}\quad \frac{\Gamma \vdash A : \mathsf{U} \qquad \Gamma, x{:}A \vdash f^+\ x = g^+\ x : B \qquad \Gamma \vdash f : \Pi(x{:}A)B \qquad \Gamma \vdash g : \Pi(x{:}A)B}{\Gamma \vdash f = g : \Pi(x{:}A)B}$$

$$\textsc{App-Fun}\quad \frac{\Gamma \vdash A : \mathsf{U} \qquad \Gamma, x{:}A \vdash B : \mathsf{U} \qquad \Gamma \vdash f = f' : \Pi(x{:}A)B \qquad \Gamma \vdash a : A}{\Gamma \vdash f\ a = f'\ a : B[a]}$$

$$\textsc{App-Arg}\quad \frac{\Gamma \vdash A : \mathsf{U} \qquad \Gamma, x{:}A \vdash B : \mathsf{U} \qquad \Gamma \vdash f : \Pi(x{:}A)B \qquad \Gamma \vdash a = a' : A}{\Gamma \vdash f\ a = f\ a' : B[a]}$$

5 Head reduction

We define head reduction $M \rightsquigarrow N$ as the compatible closure of β-reduction at the head:

$$\textsc{HR-Beta}\quad \frac{}{(\lambda(x{:}A)M)\ a \rightsquigarrow M[a]} \qquad \textsc{HR-App}\quad \frac{M \rightsquigarrow M'}{M\ a \rightsquigarrow M'\ a}$$

6 Main result: Pi injectivity

Theorem 1 (Corollary 6; Coquand–Huber 2018, p. 661)**.** *If* $\Gamma \vdash A_0 = \Pi(x{:}B_1)F_1 : \mathsf{U}$, *then there exist* B_0, F_0 *such that*

1. $A_0 \rightsquigarrow^* \Pi(x{:}B_0)F_0$,
2. $\Gamma \vdash B_0 = B_1 : \mathsf{U}$,
3. $\Gamma, x{:}B_0 \vdash F_0 = F_1 : \mathsf{U}$.

Corollary 2 (Pi–Pi injectivity)**.** *If* $\Gamma \vdash \Pi(x{:}A_0)B_0 = \Pi(x{:}A_1)B_1 : \mathsf{U}$, *then*

$$\Gamma \vdash A_0 = A_1 : \mathsf{U} \qquad \textit{and} \qquad \Gamma, x{:}A_0 \vdash B_0 = B_1 : \mathsf{U}.$$

Proof outline (Theorem). The proof proceeds by domain-theoretic adequacy.

1. From $\Gamma \vdash A_0 = \Pi B_1\, F_1 : \mathsf{U}$, extract $\Gamma \vdash A_0 : \mathsf{U}$ by presupposition (SubstitutionLemma).

2. Apply the soundness theorem (convSound') at the *bottom environment* $\rho_\perp$ (mapping every variable to $\perp$) to obtain a bidirectional evaluation transfer: for every finite approximation u,

$$\mathsf{EvalRel}(\Pi B_1\, F_1, \rho_\perp, u) \implies \mathsf{EvalRel}(A_0, \rho_\perp, u).$$

3. Instantiate with $u = \mathsf{PiCode}(\perp, \mathsf{nil})$, which is coherent because $\mathsf{CoherentFunTail}(\mathsf{nil}) = \top$. The evaluation $\mathsf{EvalRel}(\Pi B_1\, F_1, \rho_\perp, \mathsf{PiCode}(\perp, \mathsf{nil}))$ is constructible trivially (domain $\perp$, empty function graph, vacuous selection body). Transfer gives $\mathsf{EvalRel}(A_0, \rho_\perp, \mathsf{PiCode}(\perp, \mathsf{nil}))$.

4. Apply the bundled equality adequacy theorem (adequacyEqSub2) with the identity substitution at $\rho_\perp$ to obtain

$$\mathsf{EqVal2}\ \Gamma\ A_0\ (\Pi B_1\, F_1)\ \mathsf{U}\ (\mathsf{PiCode}(\perp, \mathsf{nil}))\ \mathsf{UCode}.$$

5. This unfolds to $\mathsf{EqValTyPi2}$, a Σ-type containing:

   - $\mathsf{Red}\ \Gamma\ A_0\ (\Pi\, B_0\, F_0)\ \mathsf{U}$ (giving part 1),
   - $\Gamma \vdash B_0 = B_1' : \mathsf{U}$ and $\Gamma, x{:}B_0 \vdash F_0 = F_1' : \mathsf{U}$,
   - $\mathsf{Red}\ \Gamma\ (\Pi B_1\, F_1)\ (\Pi B_1'\, F_1')\ \mathsf{U}$.

   Since Π is a head normal form, Red-unique-Pi forces $B_1' = B_1$ and $F_1' = F_1$, giving parts 2 and 3.

□

Proof of Corollary. Apply the theorem to $\Gamma \vdash \Pi A_0\, B_0 = \Pi A_1\, B_1 : \mathsf{U}$. This yields B_0', F_0' with $\Pi A_0\, B_0 \rightsquigarrow^* \Pi B_0'\, F_0'$. Since Π is a head normal form, Red-unique-Pi gives $B_0' = A_0$ and $F_0' = B_0$, so the domain and codomain conversions become $\Gamma \vdash A_0 = A_1 : \mathsf{U}$ and $\Gamma, x{:}A_0 \vdash B_0 = B_1 : \mathsf{U}$. □

Corollary 3 (Eta-conversion)**.** *If* $\Gamma \vdash A : \mathsf{U}$, $\Gamma, x{:}A \vdash B : \mathsf{U}$, *and* $\Gamma \vdash f : \Pi(x{:}A)B$, *then*

$$\Gamma \vdash f = \lambda(x{:}A) f^+\, x : \Pi(x{:}A)B.$$

Proof. By FunExt it suffices to show $\Gamma, x{:}A \vdash f^+\, x = (\lambda(x{:}A) f^+\, x)^+\, x : B$. Since $(\lambda(x{:}A)M)^+ = \lambda(x{:}A^+)M^{+\ell}$ (where $+\ell$ denotes weakening under the binder), the right-hand side is a β-redex. Applying Beta and the substitution identity $M^{+\ell}[x_0] = M$ gives the result. Formalised in `EtaConversion.agda`. □

The entire proof is formalised in Agda with `--without-K --exact-split`, no `--type-in-type`, and no postulates (in the files on which `PiInjectivity.agda` depends).

References

[1] T. Coquand and S. Huber, "An adequacy theorem for dependent type theory," *Archive for Mathematical Logic*, vol. 57, pp. 649–676, 2018.